Department of Mathematics, Thapar Institute of Engineering & Technology, Patiala

UMA021 : Numerical Linear Algebra

Assignment 1

Floating Point Arithmetic and Errors

1. Compute the absolute error and relative error in approximations of x by x^*.

(a) $x = \pi,\ x^* = 22/7$

(b) $x = \sqrt{2},\ x^* = 1.414$

(c) $x = 8!,\ x^* = 39900.$

2. Find the largest interval in which x^* must lie to approximate x with relative error at most 10^{-4} for each value of x.

(a) π

(b) e

(c) $\sqrt{3}$

(d) $\sqrt[3]{7}$.

3. A rectangular parallelepiped has sides of length 3 cm, 4 cm, and 5 cm, measured to the nearest centimeter. What are the best upper and lower bounds for the volume of this parallelepiped? What are the best upper and lower bounds for the surface area?

4. Use four-digit rounding arithmetic and the formula to find the most accurate approximations to the roots of the following quadratic equations. Compute the absolute errors and relative errors.

$$\frac{1}{3}x^2 - \frac{123}{4}x + \frac{1}{6} = 0.$$

5. The derivative of $f(x) = \dfrac{1}{(1-3x^2)}$ is given by $\dfrac{6x}{(1-3x^2)^2}$. Do you expect to have difficulties evaluating this derivative at $x = 0.577$? Try it using 3- and 4-digit arithmetic with chopping.

6. Suppose two points (x_0, y_0) and (x_1, y_1) are on a straight line with $y_1 \neq y_0$. Two formulas are available to find the x-intercept of the line:

$$x = \frac{x_0 y_1 - x_1 y_0}{y_1 - y_0}, \text{ and } x = x_0 - \frac{(x_1 - x_0)y_0}{y_1 - y_0}.$$

(a) Show that both formulas are algebraically correct.

(b) Use the data $(x_0, y_0) = (1.31, 3.24)$ and $(x_1, y_1) = (1.93, 4.76)$ and three-digit rounding arithmetic to compute the x-intercept both ways. Which method is better and why?

7. Verify that the functions $f(x)$ and $g(x)$ are identical functions.

$$f(x) = 1 - \sin x, \quad g(x) = \frac{\cos^2 x}{1 + \sin x}.$$

(a) Which function should be used for computations when x is near $\pi/2$? Why?

(b) Which function should be used for computations when x is near $3\pi/2$? Why?

8. (a) Evaluate the polynomial $y = x^3 - 7x^2 + 8x - 0.35$ at $x = 1.37$ using 3-digit arithmetic with chopping. Evaluate the percent relative error.

(b) Repeat part (8a) but express y as $y = ((x-7)x+8)x - 0.35$. Evaluate the error and compare with part (8a).

9. Let $P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ be a polynomial, and let x_0 be given. Construct an algorithm to evaluate $P(x_0)$ using nested multiplication.